I Dewa Nyoman Bayu Satria Wibawa

Machine Learning & Software Systems Engineer • Selected Projects & Sample Works

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ENGINEERING FOCUS & TECHNICAL PROFILE

Machine Learning and Software Systems Engineer experienced in building end-to-end classification pipelines, real-time IoT telemetry architectures, and production cloud services. Specializes in solving severe class imbalance using adaptive synthetic resampling (ADASYN, SMOTE), training custom deep learning models without third-party black-box dependencies, and deploying low-latency web platforms that process high-frequency streaming data.

1. Autonomous Vehicle Sentiment Analysis Platform

Organization: National Research and Innovation Agency (BRIN) • **Role:** Machine Learning Engineer • **Stack:** Python, PyTorch, ADASYN, FastAPI, Next.js, Vercel Edge • **URL:** https://brin-project-kp.vercel.app/

Engineered an end-to-end sentiment intelligence platform for BRIN to evaluate Indonesian public discourse on autonomous electric vehicles. The platform ingests social opinion streams, addresses high class imbalance, and performs real-time classification to inform national transport policy.

- **Dataset Engineering & Algorithmic Resampling:** Curated 2,000+ social commentary samples. Implemented ADASYN and SMOTE oversampling to generate synthetic minority samples along decision boundaries, rectifying heavy skew without semantic distortion.
- **Custom Neural Classifier (94.2% Accuracy):** Trained an in-house deep learning neural network on domain token embeddings, achieving **94.2% test accuracy** and an **F1-score of 0.93**, operating fully independently without costly third-party commercial APIs.
- **Production REST Inference Serving:** Containerized the inference pipeline into an asynchronous REST API, achieving deterministic response latencies under **50ms** with interactive class probability distributions.

Key Metrics: 94.2% Test Accuracy • 0.93 F1-Score • < 50ms Inference Latency • 2,000+ Curated Data Points

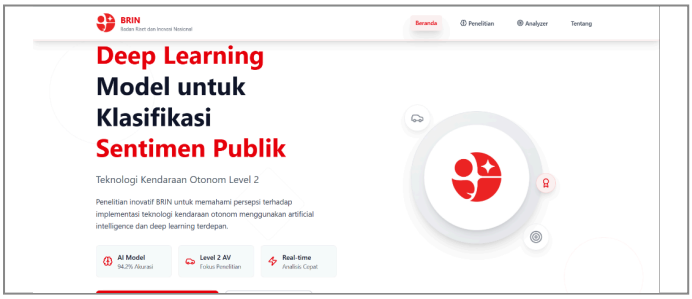


Figure 1.1: BRIN Research Portal. Interface displaying public opinion corpus navigation, dataset distribution metrics, and sentiment research documentation.

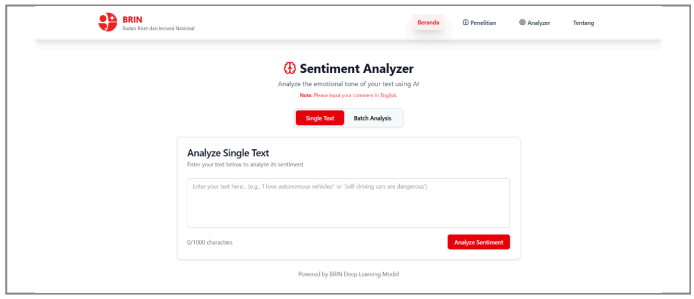


Figure 1.2: Real-Time Inference Analyzer. Live sentence-level sentiment classification demonstrating probability scoring (Positive/Neutral/Negative) and confidence metrics.

2. BySense: Photovoltaic & Solar IoT Telemetry Network

Client: PT Batari Energy (2026) • **Role:** Full-Stack & IoT Systems Engineer • **Stack:** React, TypeScript, Leaflet, Open-Meteo, Mobile APK • **URL:** https://bysense.batarienergy.com/

Engineered an industrial-grade IoT telemetry platform for PT Batari Energy to monitor distributed commercial solar photovoltaic (PV) installations nationwide. The platform provides real-time visibility into power production, storage health, and weather impact, eliminating manual inspections.

- **Nationwide Ingestion & Geospatial Telemetry:** Aggregated multi-site inverter gateways and Battery Energy Storage Systems (BESS), projecting live health indicators, active power generation, and faults across Indonesian regions onto an interactive map.
- **High-Frequency Instrumentation (10.97 kW Yield):** Implemented 5-second polling telemetry displaying active solar generation (**10.97 kW**), battery state-of-charge (**65% SoC**), building load consumption, and net grid export.
- **Irradiance-to-Power Analytical Modeling:** Built an empirical regression model correlating incident pyranometer solar irradiance (W/m^2) directly with inverter electrical output (kW) to automatically flag solar panel soiling and inverter clipping.

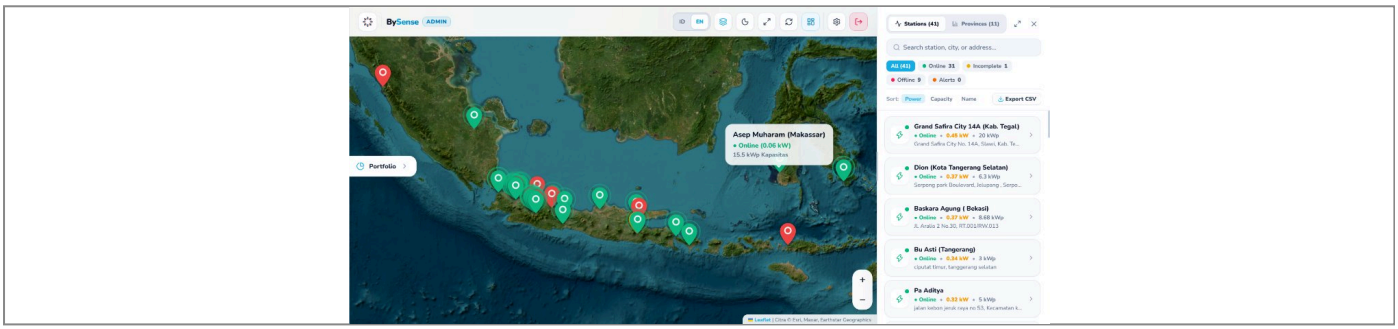


Figure 2.1: Nationwide Interactive Photovoltaic Telemetry Map. Centralized GIS monitoring view displaying geographical station pins, live operational status indicators, and active solar generation outputs across Indonesia.

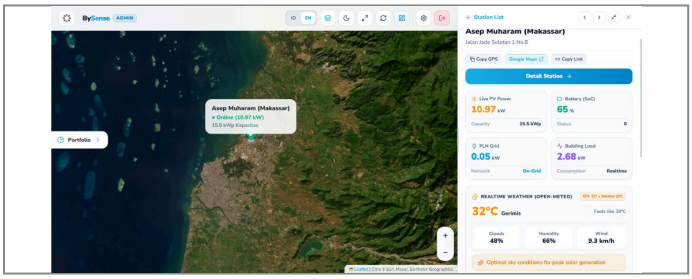


Figure 2.2: Station Telemetry Detail. High-frequency instrument view tracking 10.97 kW solar output, 65% battery SoC, building load, and daily energy total.

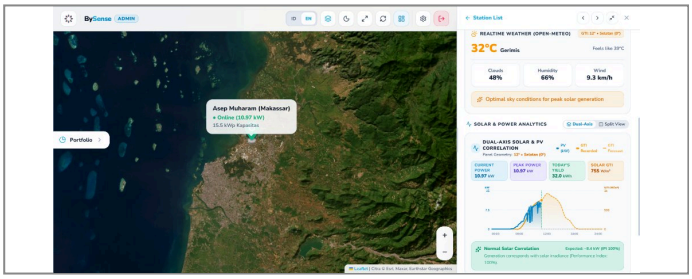


Figure 2.3: Irradiance Correlation Curve. Analytical model comparing solar irradiance (W/m^2) against kilowatt delivery to identify panel shading and hardware anomalies.

BySense Telemetry: Diurnal Tracking, Weather Sync & Mobile Release

To ensure field resilience and prevent erroneous hardware fault dispatches, BySense incorporates continuous diurnal curve tracking, meteorological telemetry synchronization, and a dedicated mobile client deployed to field engineers.

- **Diurnal Generation Curve Analysis:** Plots non-linear daylight solar curves illustrating early-morning ramp-up, peak midday yield, and late-afternoon tapering, enabling plant managers to track performance ratio (PR) indices.
- **Meteorological Synchronization via Open-Meteo:** Integrated external REST weather telemetry synchronizing localized cloud cover percentages, ambient temperature, and humidity to distinguish between cloudy weather and actual hardware degradation.
- **Mobile Client on Google Play Store:** Built and released the native Android application to the **Google Play Store**, enabling field engineers to receive automated anomaly alerts and verify station statuses on-site.



Figure 2.4: Diurnal Generation Curve. Continuous solar generation curve tracking morning ramp-up, peak midday yield, and inverter operational boundaries.

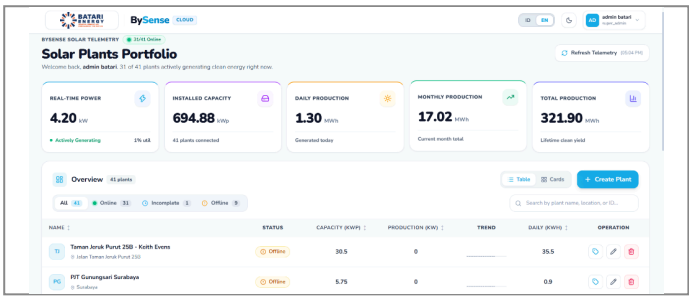


Figure 2.5: Ambient Weather Telemetry. Real-time integration of ambient temperature and cloud cover via Open-Meteo API, preventing false maintenance alerts.

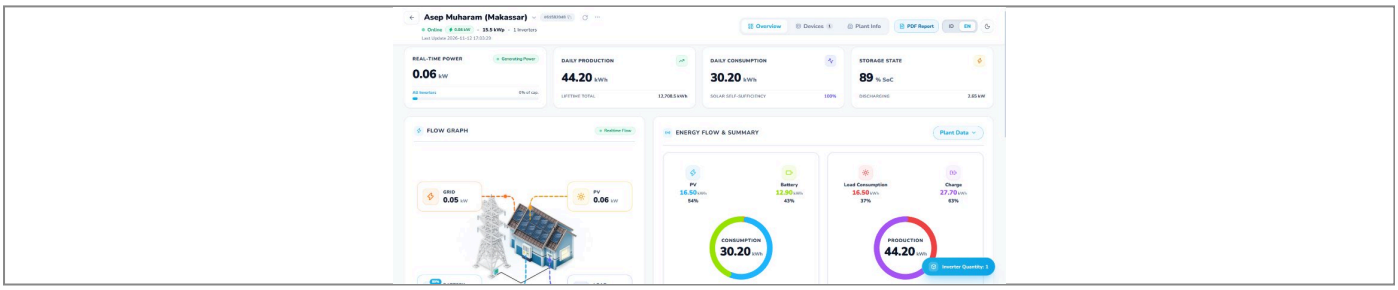


Figure 2.6: Operational Performance Dashboard. Summary view of cumulative kilowatt-hours generated, system performance ratio, grid export balance, and operational stability indices.

3. NutriMood: Multi-Criteria Nutritional Recommender

Context: DBS Foundation Coding Camp (Distinction) • **Role:** Machine Learning & Full-Stack Developer • **Stack:** Python, Scikit-learn, React, REST API • **URL:** https://nutrimood.vercel.app/

Engineered a multi-criteria dietary recommendation system matching user metabolic health constraints, caloric targets, and emotional state across a database of **1,200+ Indonesian food profiles**. Implemented hybrid filtering to output personalized culinary options with balanced macronutrient breakdowns.

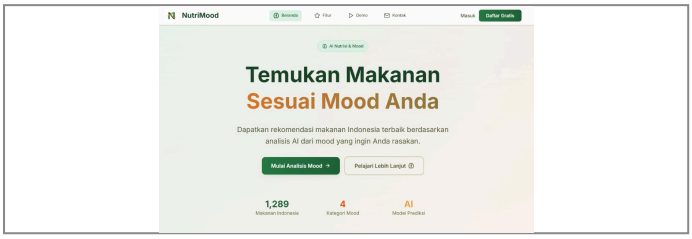


Figure 3.1: NutriMood Landing View. Health profiling, daily objective configuration, and wellness onboarding interface.

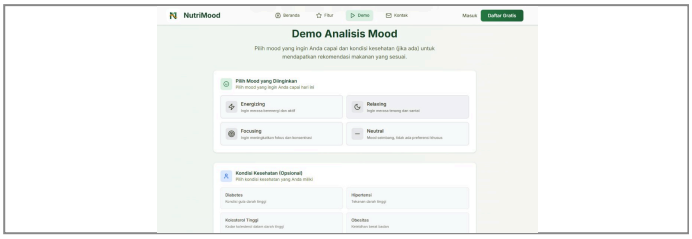


Figure 3.2: Recommender Evaluation. Interactive output view presenting matched culinary suggestions and nutritional distributions.

4. Cimahi Battery: Industrial Commercial Web Platform

Client: PT Cimahi Battery Indonesia (2026) • **Role:** Frontend & Web Systems Engineer • **Stack:** React, Tailwind CSS, Parametric Query Engine • **URL:** https://cimahibattery.com/

Architected the enterprise web platform for PT Cimahi Battery Indonesia, an industrial manufacturer of lithium iron phosphate (LiFePO4) battery packs. Implemented a parametric specification filter enabling commercial engineers to query prismatic cells by voltage, capacity, and discharge rates.

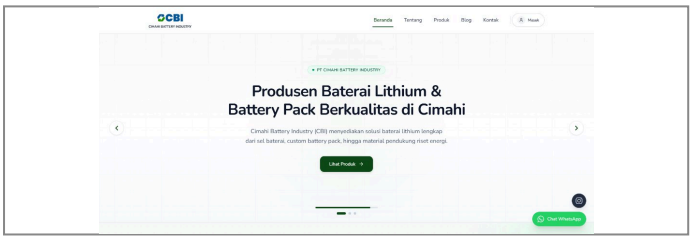


Figure 4.1: Enterprise Web Platform. Official corporate portal highlighting commercial energy storage systems and engineering certifications.

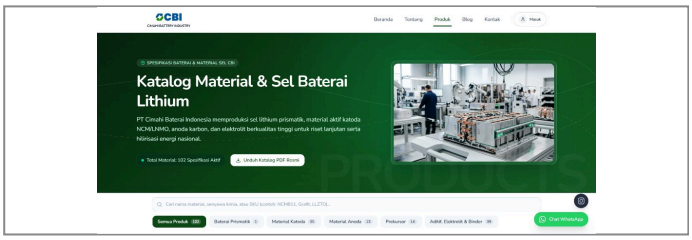


Figure 4.2: Prismatic Cell Directory. Interactive catalog enabling parametric technical filtering by voltage, capacity, cycle life, and form factor.

Engineering Authorship & Verification: All algorithms, telemetry interfaces, and web applications documented in this portfolio were engineered and verified in production by I Dewa Nyoman Bayu Satria Wibawa. Source repositories and live demonstrations are verifiable at: **BRIN:** `brin-project-kp.vercel.app` • **BySense:** `bysense.batarienergy.com` • **Portfolio:** `idewanyomanbayusw.vercel.app`